

Sonet Rajkumar

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Professional Summary

IT professional with 15+ years of experience across Telecom and Capital Markets Algorithmic Trading. Strong background in C/C++ Linux, SMSC/IMS/SIP systems, Femto/small cells, and high-frequency trading platforms for NSE/BSE. Strong in Software Development SDLC (Agile, WaterFall), debugging, performance optimisation, client interaction, and cross-functional collaboration.

Core Skills

- C, C++, C++11/14, Data Structures, Multithreading, STL
- Telecom: SS7 MAP, IPSM-GW, TCP/UDP, SIP, RTP/RTCP, SMPP, SNMP, TR-069
- Trading Tech: Algo Trading, HFT, FIX Protocol
- Tools: gdb, gcc, g++, valgrind, Git, Github, SVN, ClearCase, Wireshark, JIRA
- Systems: Linux (RHEL/CentOS/Ubuntu), Windows
- Scripting: Shell, Python

Experience

- Lead Software Engineer – AlgoMethods Software (Nov 2022 – Nov 2025)
- Senior Member Technical Staff – III (R&D) – Mavenir (Oct 2018 – Oct 2022)
- Technical Lead – Tech Mahindra (Jan 2016 – Sep 2018)
- Senior Software Engineer – Aricent/Altran/Capgemini (Mar 2012 – Dec 2015)
- Senior Engineer – Omnesys/Reinitiv (Mar 2009 – Mar 2012)

Key Projects

Project 1 — Volant / Swastika iTrade – Algo Trading Platform

Role: Lead Software Engineer

Tech: C++11/14, Linux/Windows, gcc, gdb, Git, JIRA, OMS, NNF Protocol

- *Led development of the Volant Algo Trading platform using modern C++ (11/14), implementing low-latency order processing, validation, and exchange communication workFlows aligned with NSE specifications.*
- *Enhanced and maintained the Swastika iTrade legacy C codebase, delivering CR changes, performance improvements, and defect Fixes for high-volume trading scenarios.*
- *Performed hands-on implementation, unit testing, gdb debugging, and refactoring of critical trading modules to meet latency, stability, and throughput requirements.*

- *Mentored junior engineers, performed code and documentation reviews, and ensured conformance with exchange protocol guidelines and software quality standards.*

Project 2 — Comverse ECS – SMSC / IPSM-GW

Role: Technical Team Lead

Tech: C++, Linux/Unix, gcc/gdb, ClearCase, DiME, MGTS, JIRA, SIP, RTP/RTCP, MAP, SMPP

- *Led the development of core C++ components within Comverse's IMS-based SMSC/ECS platform, delivering carrier-grade messaging and IP-communication services for high-volume telecom networks.*
- *Collaborated with global architects and stakeholders to analyse requirements, define architecture, and review CR's for enhancements across SMSC and IMS modules.*
- *Engineered scalable backend features including routing logic, store-and-forward modules, retry algorithms, and integrated multiple telecom and IP communication protocols such as TCP/UDP, MAP, SMPP, SIP, and RTP/RTCP.*
- *Executed hands-on C++ development, performance profiling, gdb debugging, and production-grade troubleshooting for real-time, low-latency messaging systems.*
- *Leading the India Bangalore C++ Dev team, oversaw Agile sprints and conducted code/documentation reviews, ensuring high engineering standards across cross-location teams in Israel, the US, and India.*

Project 3 — NSN Femto Gateway (FAP Startup & Communications)

Role: Senior Software Engineer

Tech: C++, Linux, gdb, valgrind, TTCN, MySQL, XML, XSD, Shell Scripts

- *Assessed, administered, and enhanced Femto Access Point (FAP) startup and communication modules for NSN's Femto Gateway solution based on 3G protocols and IP-ATM proxy Flows.*
- *Improved interoperability across legacy core networks and Home Node-B components by optimising gateway communication paths and validating protocol behaviour.*
- *Conducted performance testing, memory profiling (valgrind), TTCN-based protocol verification, and system optimisation to meet 3GPP quality standards.*
- *Worked as part of the Scrum & Sprints Delivery Team to support MT & CI runs, resolving defects and delivering feature enhancements.*
- *Performed in-depth debugging, MySQL data analysis, and XML/XSD updates while collaborating with cross-functional teams during gateway deployment and integration cycles.*

Project 4 — Service Provisioning Platform (SPP) – PoC

Role: Senior Software Engineer

Tech: Linux/Windows, C/C++, WebLogic, BlackDuck Protex, Eclipse, Vi, gcc

- *Analysed and validated FOSS usage within the Service Provisioning Platform (SPP) to ensure full compliance with open-source licensing via BlackDuck Protex.*

- *Identified license risks, non-compliant components, and remediation approaches to enable safe, policy-aligned deployment.*
- *Delivered code updates and technical solutions addressing FOSS violations for multi-protocol provisioning workflows integrated with network elements.*
- *Collaborated with architects and network teams to validate provisioning flows, service mapping logic, and interoperability during PoC execution.*
- *Produced compliance documentation, analysis reports, and updated code modules to support FOSS adherence across the provisioning stack.*

Project 5 — NEST Strategy – Automated Trading System

Role: Software Engineer

Tech: Linux/Windows, C/C++, Shell Scripts, Altova MapForce, Eclipse, Vi, gcc

- *Contributed to Omnesys NEST Strategy, an automated trading system enabling programmable strategies for cash, futures, and options across NSE/BSE/MCX/NCDEX exchanges.*
- *Developed backend logic for strategy parsing, validation, and execution to support automated order generation and arbitrage workflows.*
- *Collaborated with trading SMEs to translate strategy rules into optimised backend modules compliant with exchange protocols.*
- *Supported integration testing, debugging, and optimisation of strategy scripts for reliability and accuracy in production environments.*

Education

B.E. (Information Science), VCET Puttur – VTU Belgaum (2004–2008)

Certifications, Courses & Training

- *Completed internal Mavenir L&D's hands on training on usage of Kubernetes and Docker.*
- *Completed internal Aricent L&D's training on 3G, 4G and UMTS Telecom technology.*
- *Certified Course in Web Component Development using Java Technologies from NIIT & Sun Microsystems.*
- *Certified Course in Networking & Router configuration from Everonn-Mercury Software.*

Achievements & Awards

- *ACE WOW Award (twice) – Aricent Customer Excellence Award*
- *Certificate of Recognition – for BlackDuck Protex PoC in Aricent SPARK Event*
- *Certificate of Completion for participation in Aricent Bangalore HACK100.*